

A Voyage to the Kernel



Part 20

Segment: 3.9, Day 19

In the last segment, we covered some of the basic aspects related to Linux networking. Some readers asked why they were unable to locate a few files specific to networking. You need to be aware that the location of files varies between distros. Your distro's doc/help files will help you find them. For your reference, I am including Table 1 that gives the location of a few important files in different distros.

Here is the shell script that performs the boot actions in my Debian-based distro (this file is located at `/etc/init.d/networking`):

```
#!/bin/sh -e
PATH="/usr/local/sbin:/usr/local/bin:/sbin:/bin:/usr/sbin:/usr/bin"
[ -x /sbin/ifup ] || exit 0
./lib/lsb/init-functions

# helper function to set the usplash timeout, https://launchpad.net/bugs/21617
usplash_timeout () {
    TIMEOUT=$1
    if [ -x /sbin/usplash_write ]; then
        /sbin/usplash_write "TIMEOUT $TIMEOUT" || true
    fi
}

process_options() {
    [ -e /etc/network/options ] || return 0
```

```
log_warning_msg "/etc/network/options still exists and it will be IGNORED! Read README.Debian of netbase."
}
```

```
check_network_file_systems() {
    [ -e /proc/mounts ] || return 0
```

```
exec 0<&0 < /proc/mounts
while read DEV MPTP FSTYPE REST; do
    case $DEV in
        /dev/hib*|/dev/nd[a-z]*|/dev/etherd/*)
            log_warning_msg "not deconfiguring network interfaces:
network devices still mounted."
            exit 0
            ;;
        esac
        case $FSTYPE in
            nfs|nfs4|usbfs|nfs|nfs4|cifs|coda|ocfs2|gfs|pvfs|pvfs2|
fuse.httpfs|fuse.curittps)
            log_warning_msg "not deconfiguring network interfaces:
network file systems still mounted."
            exit 0
            ;;
        esac
    done
exec 0<&9 9<&8-
}
```

.....code truncated

force-reload|restart)

Distro	Configuration	Boot specific ones
Debian*	/etc/init.d/network	/etc/rc2.d/
Red-Hat	/etc/rc.d/init.d/network	/etc/rc3.d/
Slackware	/etc/rc.d/rc.inet1	/etc/rc.d/rc.inet2

(*there could be minor changes in Debian-based distros)

Table 1